

Higher-Order Metaphysics
Week Two: higher-order logic languages
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1 Recap: the lambda-language of a signature

2 Substitution

We define a syntactic operation $[N/a]$ for every variable or constant a and term N of the same type, which replaces all **free** occurrences of a with occurrences of N .¹ This can be defined recursively as follows:

$$\begin{aligned} a[N/a] &= N \\ u[N/a] &= u && \text{when } u \text{ is a variable other than } a \\ c[N/a] &= c && \text{when } c \text{ is a constant other than } a \\ (BC)[N/a] &= (B[N/a]C[N/a]) \\ (\lambda v.B)[N/v] &= (\lambda v.B) \\ (\lambda u.B)[N/a] &= (\lambda u.B[N/a]) && \text{when } u \text{ is a variable other than } a \end{aligned}$$

¹ Bacon goes for an interesting non-standard definition of substitution, but the version of the book we have contains several confusing errors in its definition, so I'm taking the trouble to explain a more standard one.

Sometimes, substituting N for free occurrences of a in a term leads to *variable capture*, where occurrences of free variables in N get bound. For example, $(\lambda x.Rxy)a[x/y]$ is $(\lambda x.Rxx)a$. We don't like it when this happens! Let's say that N is **free for** variable or constant a in a term B iff all of the free variable occurrences in N are still free in $B[N/a]$ —I'll write down a rigorous definition of this in the margin, but the important thing is to have an intuitive understanding.²

3 The relation of immediate β -equivalence

Definition. A **immediately β -reduces** to B iff for some M, N, v , where N is free for v in M , A (the 'redex') is $((\lambda v.M)N)$ and B (the 'reduct') is $M[N/v]$.

A and B are **immediately β -equivalent** iff A immediately β -reduces to B or B immediately β -reduces to A .

According to Bacon, immediately beta-equivalent terms are *synonymous* insofar as they are intelligible at all. This seems too strong to me—synonymy is always relative to a linguistic community or practice, and we can imagine all sorts of weird and wonderful practices that would

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- N is free for a in any variable or constant.
- N is free for a in (BC) if it's free for a in both B and C .
- N is free for a in $(\lambda v.B)$ iff EITHER v isn't free in B OR it's free for a in B and v isn't in $FV(B)$.

If you prefer Bacon's more permissive notion of substitution that targets both bound and free variable-occurrences, delete the first disjunct from the second bullet point.

confer meanings on the terms of some (syntactically specified) lambda-language, and confer different meanings on immediately β -equivalent terms. Still it might be true that there is a “standard” range of ways of using lambda-languages that makes the synonymy claim true.³

Be that as it may, the relation of immediate β -equivalence will play an important role in any reasonable logic for a language containing lambdas.

4 The relation of immediate α -equivalence

Definition. A is **immediately α -equivalent** to B iff A is $\lambda v.M$ and B is $\lambda u.M[u/v]$ for some variable u that isn’t free in M .⁴

Typical α -equivalent terms are $\lambda x.Rxx$ and $\lambda y.Ryy$ (where R is a constant). They are “mere alphabetic variants” that differ only in which variable we choose to use. On the natural ways of understanding lambda-languages, it seems *very* plausible that immediately α -equivalent terms are synonymous!

5 The relation of β -equivalence

Speaking a bit informally, the idea is that A and B are β -equivalent iff we can turn A into B by a succession of operations where at each step we substitute an occurrence of a term (maybe an embedded one) for an occurrence of some immediately β -equivalent or immediately α -equivalent term.

To characterize this rigorously, we can introduce the following independently useful syntactic notion:

Definition. A relation $\sim$ on the terms of some lambda-language is a congruence iff it is reflexive, symmetric, and transitive, $(AB) \sim (CD)$ whenever $A \sim C$ and $B \sim D$, and $(\lambda v.A) \sim (\lambda v.B)$ whenever $A \sim B$.

We can then say what we mean quite succinctly:

Definition. **β -equivalence** is the smallest congruence containing immediate β -equivalence and immediate α -equivalence.

Synonymy as we are understanding it is presumably a congruence, so we’re using the language in a way that makes immediately β - and α -equivalent terms synonymous, β -equivalent terms will also be synonymous.

³ Notice that if we hadn’t include the “free for” proviso in the definition, the synonymy claim would be completely hopeless— $(\lambda yx.Rxy)xy$ would then be counted as immediately β -equivalent to $(\lambda x.Rxx)y$ which is in turn immediately β -equivalent (even on our definition) to Ryy ; but $(\lambda yx.Rxy)xy$ is not plausibly synonymous with Ryy !

⁴ Note that if this relation obtains, A is $\lambda v.M[u/v][v/u]$, so it obtains in both directions—there is no natural notion of “immediate α -reduction” that stands to immediate α -equivalence as immediate β -reduction stands to immediate β -equivalence.

6 The relation of immediate η -equivalence

There is another relation which plays a similar role in the logics we'll be focusing on, though it is less crucial.

Definition. A **immediately η -reduces** to B iff A (the η -redex) is $\lambda v.Bv$ for some variable v that isn't free in B (the η -reduct).⁵ A is **immediately η -equivalent** to B iff A immediately η -reduces to B or B immediately η -reduces to A .

⁵ Note that for this to make sense, B must be of some type $\sigma \rightarrow \tau$ where σ is the type of v , and hence so must A .

The fact that our syntax lets us make terms like $(\lambda v.Bv)$ (where v isn't free in B) seems like a sort of edge case—the point of having lambdas in the language is to let us build interesting terms like $\lambda xy.Ryx$ and $\lambda x.Rxx$, but to avoid making arbitrary restrictions we then have to let in η -redexes even though they don't seem to do anything interesting. This supports the idea that they will end up synonymous with their reducts, and that we should be looking for a logic that licenses such substitutions.

Note that if B is a lambda-term $\lambda u.C$, $(\lambda v.Bv)$ is β -equivalent to B , since Bv is immediately β -equivalent to $C[v/u]$, and $\lambda v.C[v/u]$ is immediately α -equivalent to $\lambda u.C$, i.e. B . So the only “new” immediate η -equivalences are those where B is a variable or constant.

7 The relation of $\beta\eta$ -equivalence

Definition. $\beta\eta$ -equivalence is the smallest congruence containing immediate β -equivalence and immediate η -equivalence.

Note that we didn't bother to throw in immediate α -equivalence in there—that's because we get it for free. If u isn't free in M , $(\lambda v.M)$ is immediately η -equivalent to $\lambda u.(\lambda v.M)u$, and $(\lambda v.M)u$ is immediately β -equivalent to $M[u/v]$, so $(\lambda v.M)$ is $\beta\eta$ -equivalent to $\lambda u.M[u/v]$.

8 The signature of higher-order logic

Λ is the following signature:

$$\begin{aligned}
 & \top, \perp : t \\
 & \neg : t \rightarrow t \\
 & \wedge, \vee, \rightarrow, \leftrightarrow : t \rightarrow t \rightarrow t \\
 & \forall_\sigma, \exists_\sigma : (\sigma \rightarrow t) \rightarrow t & \text{(for each type } \sigma) \\
 & =_\sigma : \sigma \rightarrow \sigma \rightarrow t & \text{(for each type } \sigma)
 \end{aligned}$$

$\mathcal{L}(\Lambda)$ is the higher-order language with exactly these constants. A *higher-order* language is $\mathcal{L}(\Lambda \cup \Sigma)$ for some signature Σ (of ‘nonlogical constants’).

We write $\wedge, \vee, \rightarrow, \leftrightarrow, =_\sigma$ in infix position, and write $\forall v.P$ and $\exists v.P$ for $\forall_\sigma(\lambda v.P)$ and $\exists_\sigma(\lambda v.P)$ where σ is the type of v .

We’ll also be interested in smaller signatures, in which case we may treat some of the above symbols not as constants but as abbreviations of complex terms built from our favoured constants.

9 Axiomatic and natural deduction presentations of a logic

There are several different ways of specifying a “logic” for a given language. In a *Hilbert-style* or *axiomatic* approach, one identifies the logic with its set of *theorems* (which are formulae, i.e. type- t terms), which one specifies as the closure of a certain set of syntactically *axioms* under a certain (usually very small) set of *inference rules* (which are relations between a nonempty finite set of formulae and a formula).

In a (“sequent-style”) *natural deduction* approach, one identifies the logic with its *provability* or *consequence* relation, which is a set of *sequents*, each of them an ordered pair $\langle \Gamma, \varphi \rangle$ (also notated $\Gamma \triangleright \varphi$) where Γ is a finite set of formulae and φ is a formula, which one specifies as the closure of a certain (usually small) set of *axiom-sequents* under a certain set of *inference rules* (which are relations between a nonempty finite set of sequents and a sequent).⁶

The advantage of the axiomatic approach is that it’s transparent what philosophically interesting commitment is involved in “endorsing” a logic (for a language you understand): you’re presumably saying at least that all of its closed theorems are *true*.⁷

Natural-deduction approaches offer advantages of elegance and familiarity (since intro logic classes normally teach natural deduction). A disadvantage is that it may not be transparent what good status a “proponent” of a certain logic is claiming for the sequents that belong to it (since it is not obvious what it would mean for a sequent to be “true”) so there is more danger of getting bogged down in debates about analyticity/necessity/whatever. But minimally, these “proponents” will presumably think that whenever the logic contains a sequent of the form $\emptyset \triangleright \varphi$ and φ is closed, φ is true; and that’s enough to be going on with. We are free to treat all the other sequents as mere “plumbing” for generating these ones, just as we might treat open theorems on an axiomatic approach.

⁶ Alternatively, we could set things up so that Γ is a *multiset* or *list* of formulae. For some logics that don’t include classical propositional logic, this is crucial; in the classical case, it depends on what we find most elegant.

⁷ A lover of a certain logic *might* happen to think that its theorems have some further status like analyticity or necessity, but all the worries about logics we’ll be addressing with in this course are worries already about the truth claim and not these further claims.

10 Natural deduction presentation of H

I'm going to start with a basic system of logic called H , which can be given in both natural deduction and axiomatic ways; later we'll look at some interesting variants (including weakenings and strengthenings).

Let's begin with a logic for the smaller signature $\{\rightarrow, \neg, \perp, \forall_\sigma\}$.

Definition. For a signature Σ , the *classical higher order logic* $H(\Sigma)$ is the smallest relation $\vdash$ between finite subsets of $\mathcal{L}(\Sigma)^t$ and members of $\mathcal{L}(\Sigma)^t$ meeting the following conditions:⁸

$$\begin{array}{ll}
 \frac{P \in \Gamma}{\Gamma \vdash P} \text{ Assumption} & \frac{\Gamma \vdash P \quad P \sim_{\beta\eta} Q}{\Gamma \vdash Q} \text{ conv} \\
 \frac{\Gamma, P \vdash Q}{\Gamma \vdash P \rightarrow Q} \rightarrow\text{Intro} & \frac{\Gamma \vdash P \rightarrow Q \quad \Gamma \vdash P}{\Gamma \vdash Q} \rightarrow\text{Elim} \\
 \frac{\Gamma \vdash P \quad v \notin FV(\Gamma)}{\Gamma \vdash \forall v.P} \forall\text{-intro} & \frac{\Gamma \vdash \forall_\sigma F \quad A : \sigma}{\Gamma \vdash FA} \forall\text{-Elim} \\
 \frac{\Gamma, P \vdash \perp}{\Gamma \vdash \neg P} \neg\text{-Intro} & \frac{\Gamma \vdash P \quad \Gamma \vdash \neg P}{\Gamma \vdash \perp} \neg\text{-Elim} \\
 \frac{\Gamma \vdash \neg(\neg P)}{\Gamma \vdash P} \text{ DNE} & \frac{\Gamma \vdash \perp \quad P : t}{\Gamma \vdash P} \perp\text{Elim}
 \end{array}$$

⁸ The horizontal lines here are just short for 'if, then' sentences in our metalanguage (standard mathematical English). This should be distinguished from a confusingly different use of horizontal lines in specifying certain syntactic objects (that are trees as opposed to strings).

The hallmark here—and the only interesting departure from standard classical first order logic for languages without lambda—is the conversion rule. The rules for $\rightarrow, \neg, \neg$ are just the standard rules from classical propositional logic. The $\forall$ -Intro rule is just the generalization of the familiar one to arbitrary types. The $\forall$ -Elim rule may look slightly unfamiliar, but in the presence of the conversion rule it could equally well be given as

$$\frac{\Gamma \vdash \forall_\sigma v.P \quad A : \sigma}{\Gamma \vdash P[A/v]}$$

with the requirement that A be free for v in P .⁹

Here are two key properties of this H , both of which can be established by easy inductions, and will carry over to all systems we will consider.

$$\begin{array}{ll}
 \frac{\Gamma \vdash P \quad \Gamma \subseteq \Gamma'}{\Gamma' \vdash P} \text{ Weakening} & \frac{\Gamma \vdash P \quad B \text{ free for } a \text{ in } \Gamma \text{ and } P}{\Gamma[B/a] \vdash P[B/a]} \text{ Substitution}
 \end{array}$$

⁹ So long as that requirement is met, $(\lambda v.P)A$ is β -equivalent to $P[A/v]$. In the other direction, F is immediately η -equivalent to $\lambda v.Fv$ (if F is a variable or constant) and β -equivalent to it if F is a lambda term, so $\forall_\sigma F$ is $\beta\eta$ -equivalent to $\forall v.Fv$.

11 Adding the remaining logical constants

For future reference, the following are the standard natural deduction rules for the remaining logical constants, which we'll need to add if we are adding them as primitive.

$$\begin{array}{c}
\frac{\Gamma \vdash P \quad \Gamma \vdash Q}{\Gamma \vdash P \wedge Q} \wedge\text{Intro} \qquad \frac{\Gamma \vdash P \wedge Q}{\Gamma \vdash P} \wedge\text{Elim1} \qquad \frac{\Gamma \vdash P \wedge Q}{\Gamma \vdash Q} \wedge\text{Elim2} \\
\\
\frac{\Gamma \vdash P}{\Gamma \vdash P \vee Q} \vee\text{Intro1} \qquad \frac{\Gamma \vdash P}{\Gamma \vdash Q \vee P} \vee\text{Intro2} \qquad \frac{\Gamma \vdash P \vee Q \quad \Gamma, P \vdash R \quad \Gamma, Q \vdash R}{\Gamma \vdash R} \vee\text{-Elim} \\
\\
\frac{\Gamma, P \vdash Q \quad \Gamma, Q \vdash P}{\Gamma \vdash P \leftrightarrow Q} \leftrightarrow\text{Intro} \qquad \frac{\Gamma \vdash P \leftrightarrow Q \quad \Gamma \vdash P}{\Gamma \vdash Q} \leftrightarrow\text{Elim1} \qquad \frac{\Gamma \vdash P \leftrightarrow Q \quad \Gamma \vdash Q}{\Gamma \vdash P} \leftrightarrow\text{Elim2} \\
\\
\frac{}{\Gamma \vdash \top} \top\text{-Intro} \\
\\
\frac{\Gamma \vdash FA}{\Gamma \vdash \exists_\sigma F} \exists\text{Intro} \qquad \frac{\Gamma \vdash \exists v.P \quad \Gamma, P[u/v] \vdash Q \quad u \notin FV(\Gamma)}{\Gamma \vdash Q} \exists\text{Elim} \\
\\
\frac{A : \sigma}{\Gamma \vdash A =_\sigma A} =\text{Intro} \qquad \frac{\Gamma \vdash A =_\sigma B \quad \Gamma \vdash FA}{\Gamma \vdash FB} =\text{Elim}
\end{array}$$

If we *don't* want to take some or all of these symbols as primitive, there are ways of defining them in terms of $\rightarrow, \neg, \perp, \forall_\sigma$ for which the rules above can be derived from the ones in the previous section. Bacon (p. ****) gives one set of definitions with this property; here is a slightly different set that is less intuitive but which has the advantage that it's easier to show that the rules for the defined constants are admissible, and does not require the use of the DNE rule which intuitionists reject.

$$\begin{aligned}
\wedge &:= \lambda pq. \forall r. (p \rightarrow q \rightarrow r) \rightarrow r \\
\vee &:= \lambda pq. \forall r. (p \rightarrow r) \rightarrow (q \rightarrow r) \rightarrow r \\
\leftrightarrow &:= \lambda pq. p \rightarrow q \wedge q \rightarrow p \\
\top &:= \neg \perp \\
\exists_\sigma &:= \lambda X^{\sigma \rightarrow t}. \forall p^t. (\forall y^\sigma. Xy \rightarrow p) \rightarrow p \\
=_\sigma &:= \lambda y^\sigma z^\sigma. \forall X^{\sigma \rightarrow t}. Xy \rightarrow Xz
\end{aligned}$$

As an example, here is how you show the admissibility of the $\wedge\text{Intro}$ rule with the above definition. Choose a variable v of type t that is not

free in Γ, P , or Q

$$\begin{array}{c}
 \frac{\Gamma \vdash P}{\Gamma, P \rightarrow Q \rightarrow v \vdash P} \text{Weakening} \quad \frac{}{\Gamma, P \rightarrow Q \rightarrow v \vdash P \rightarrow Q \rightarrow v} \text{Assumption} \quad \frac{\Gamma \vdash Q}{\Gamma, P \rightarrow Q \rightarrow v \vdash Q} \text{Weakening} \\
 \frac{}{\Gamma, P \rightarrow Q \rightarrow v \vdash Q \rightarrow v} \rightarrow\text{Elim} \quad \frac{}{\Gamma, P \rightarrow Q \rightarrow v \vdash Q} \rightarrow\text{Elim} \\
 \frac{}{\Gamma, P \rightarrow Q \rightarrow v \vdash v} \rightarrow\text{Intro} \\
 \frac{}{\Gamma \vdash (P \rightarrow Q \rightarrow v) \rightarrow v} \rightarrow\text{Intro} \\
 \frac{}{\Gamma \vdash \forall v.(P \rightarrow Q \rightarrow v) \rightarrow v} \forall\text{Intro} \\
 \frac{}{\Gamma \vdash (\lambda pq.\forall r.(p \rightarrow q \rightarrow r) \rightarrow r)PQ} \text{conv}
 \end{array}$$

Even if we add the new constants as primitive, we could still introduce $\wedge^*$, $\vee^*$, etc. as abbreviations for the defined versions given above. With the basic rules for the constants, we will be able to prove the *coextensionality* of the defined constants and the primitive ones, e.g. $\forall pq.(p \wedge q) \leftrightarrow (p \wedge^* q)$, $\forall X^{\sigma \rightarrow t}.\forall_\sigma X \leftrightarrow \forall_\sigma^* X$, $\forall xy.(x =_\sigma y) \leftrightarrow (x =_\sigma^* y)$.

But we won't be able to prove identities like $(\wedge) =_{t \rightarrow t \rightarrow t} (\wedge^*)$ —H is quite weak in its capacity to prove identities!

- In fact, it can be shown that the only formulae $A =_\sigma B$ which are theorems of H are those for which $A \sim_{\beta\eta} B$.¹⁰

¹⁰ This is proved in Fritz, Lederman, and Uzquiano 2021; it's far from straightforward.

If we have a taste for minimizing the number of undefined constants, we can go further by also defining $\neg$ and $\perp$ in terms of $\rightarrow$ and $\forall_\sigma$:

$$\begin{aligned}
 \perp &:= \forall p.p \\
 \neg &:= \lambda p.p \rightarrow \perp
 \end{aligned}$$

Given these definitions, we can drop the rules $\perp\text{Elim}$ and $\neg\text{Intro}$, which can now be derived from the remaining ones. But we can't similarly drop DNE—if we want classical rather than intuitionistic logic, there is no getting away from having it or some other rule/axiom (like the law of excluded middle) that doesn't match the usual structure of intro and elim rules.

12 Axiomatic presentation of H

Bacon, like much of the recent philosophy literature, instead uses an axiomatic format for the logic. Here H would be identified with a set of formulae rather than a set of sequents. For the signature $\{\rightarrow, \forall_\sigma\}$, with $\neg$ defined as $\lambda p.p \rightarrow \forall q.q$, the relevant set could be specified as the smallest set T that contains all instances of the following *axiom-schemas*:

$$PC1 \quad P \rightarrow Q \rightarrow P$$

PC2 $(P \rightarrow Q \rightarrow R) \rightarrow (P \rightarrow Q) \rightarrow P \rightarrow R$

PC3' $\neg\neg P \rightarrow P$

UI $\forall_\sigma F \rightarrow FA$

$\beta\eta$ $P \rightarrow Q$ where $P \sim_{\beta\eta} Q$.

and is closed under the following two rules:

MP If $P \rightarrow Q \in T$ and $P \rightarrow T$ then $Q \in T$

Gen If $P \rightarrow Q \in T$ and $v \notin FV(P)$ then $P \rightarrow \forall v.Q \in T$.

We can show that this set is exactly the set of formulae P such that the natural-deduction version contains the sequent $\emptyset \triangleright P$.

13 Weakening $\beta\eta$ -conversion

The presence of the $\beta\eta$ -conversion rule in H gives rise to some metaphysically controversial theorems, such as the following:

- $O((\lambda x.Rxb)a) \leftrightarrow O(Rab)$, where O is a constant of type $t \rightarrow t$. $\forall Z^{t \rightarrow t}. Z((\lambda x.Rxb)a) \leftrightarrow Z(Rab)$.
- $(\lambda x.Rxb)a =_t Rab$

There are various interesting philosophical arguments against these, including

- Arguments from structure: $(\lambda x.Rxb)a$ is a predication of the monadic property $(\lambda x.Rxb)$, whereas Rab is not, so they are distinct.
- Arguments from grounding: Rab grounds $(\lambda x.Rxb)a$, and nothing grounds itself, so they are distinct.
- Arguments from attitudes: Someone (e.g. someone who wasn't great at logic) could believe Rab but not $(\lambda x.Rxb)a$, so they are distinct.

We'll say more about these and other arguments later. If you thought that some such argument was good, or were just on the fence, you might like to have a weaker system of logic that avoided these controversial consequences.

But just dropping the conversion rule gives something *very* weak—we'll treat lambda terms just as if they were arbitrary new constants, with no sensitivity to their internal structure.

If we want some minimal such sensitivity, the following should not be controversial in the same way as H:

Extensional Beta $(\lambda v_1 \dots v_n. P)A_1 \dots A_n \leftrightarrow P[A_1/v_1, \dots, A_n/v_n]$, provided that $v_1, \dots, v_n$ are all distinct and each A_i is free for v_i in P .¹¹

In an axiomatic presentation, we'd add the biconditional as an axiom in place of $\beta\eta$. In a natural deduction presentation it would be more natural to present the two directions as two new rules.

¹¹ For simultaneous substitution see Bacon, ***. Note that the simultaneous substitution $P[A_1/v_1, \dots, A_n/v_n]$ is only defined when $v_1, \dots, v_n$ are distinct, so it would make no sense to drop that requirement.

References

Fritz, Peter, Lederman, Harvey, and Uzquiano, Gabriel (2021), "Closed Structure", *Journal of Philosophical Logic*, 50: 1249–91.